

## BARIS ASKIN

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### EDUCATION

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**Carnegie Mellon University, Pittsburgh, USA** *Fall 2022 - Spring 2027 (Anticipated)*

· Ph.D. in Electrical and Computer Engineering (M.Sc. en route, 2025) **GPA: 4.0/4.0**

*Advisors: Gauri Joshi and Carlee Joe-Wong*

**Bilkent University, Ankara, Turkey** *2017 - 2022*

· B.Sc. in Electrical and Electronics Engineering **GPA: 3.99/4.0**

*Valedictorian – Highest Ranked Graduate* of 2022 in the Faculty of Engineering

### RESEARCH AREAS

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LLMs, Interpretability, Alignment, Post-Training & RL (RLVR, distillation, fine-tuning), LLM Routing, Distributed/Federated Learning

### WORK & RESEARCH EXPERIENCE

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**Graduate Researcher** *August 2022 – Present*

*Carnegie Mellon University*

- LLM Alignment: emergent & subliminal misalignment analysis through training data and signal (SFT, on-/off-policy distillation) [[NeurIPS'26](#)]
- LLM Interpretability: info.-theoretic method for internal planning horizon and branch awareness [[ICLR'26](#)]
- LLM RLVR: federated GRPO with off-policy coordination on public data [[ICML'26 Workshop](#)]
- LLM Routing: learning routers from sparse, decentralized evaluations in federated settings [[NeurIPS'26](#)]
- Federated Learning and Fine-Tuning: communication-efficient fine-tuning (LoRA) and asynchronous FL with convergence guarantees [[NeurIPS'25](#)][[AISTATS'25](#)][[UAI'24](#)]

**Research Intern** *May 2025 – August 2025*

*NVIDIA Corporation*

- Addressed client staleness in large-scale async FL, where outdated updates degrade the global model
- Designed a server-side data-free knowledge distillation method to revive stale updates [[NeurIPS'26](#)]
- My proposed method was integrated into NVIDIA FLARE, NVIDIA's open-source FL framework [[Code](#)]

**Quant Researcher Intern** *June 2026 – August 2026*

*IMC Trading*

- Quantitative research on index options
- Analyzed large-scale trade data and developed statistical methods to detect patterns

### SELECTED PUBLICATIONS

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- Emergent and Subliminal Misalignment Through the Lens of Data-Mediated Transfer**  
B. Askin\*, M. Ustaomeroglu\*, A. Nayak\*, G. Joshi, G. Qu, C. Joe-Wong  
*NeurIPS 2026* [[Link](#)]
- Internal Planning in Language Models: Characterizing Horizon and Branch Awareness**  
M. Ustaomeroglu\*, B. Askin\*, G. Joshi, C. Joe-Wong, G. Qu \*equal contribution  
*ICLR 2026* [[Link](#)][[Poster](#)]
- Reviving Stale Updates: Data-Free Knowledge Distillation for Asynchronous FL**  
B. Askin, H. R. Roth, Z. Sun, C. Joe-Wong, G. Joshi, Z. Xu  
*NeurIPS 2026* [[Link](#)][[Code](#)]

4. **Federate the Router: Learning LM Routers with Sparse and Decentralized Evaluations**  
B. Askin\*, S. Patel\*, A. Nayak\*, A. Vigano, J. Woo, G. Joshi, C. Joe-Wong  
*NeurIPS 2026* [[Link](#)][[Code](#)]
5. **PubSwap: Public-Data Off-Policy Coordination for Federated RLVR**  
A. Nayak\*, B. Askin\*, M. Ustaomeroglu, C. Joe-Wong, G. Joshi  
*ICML 2026 Workshop on RL from World Feedback* [[Link](#)]
6. **Federated Communication-Efficient Multi-Objective Optimization**  
B. Askin, P. Sharma, G. Joshi, C. Joe-Wong  
*AISTATS 2025* [[Link](#)][[Code](#)][[Poster](#)]
7. **Ravan: Multi-Head Low-Rank Adaptation for Federated Fine-Tuning**  
A. Raje, B. Askin, D. Jhunjhunwala, G. Joshi  
*NeurIPS 2025* [[Link](#)]
8. **FedAST: Federated Asynchronous Simultaneous Training**  
B. Askin, P. Sharma, C. Joe-Wong, G. Joshi  
*UAI 2024* [[Link](#)][[Code](#)][[Poster](#)]
9. **Aligned Data Can Induce Misalignment via Context Confusion**  
Y. Bakman, D. N. Yaldiz, B. Askin, S. Roy, M. Ziyadi, S. Avestimehr, S. P. Karimireddy  
*Preprint 2026* [[Link](#)]
10. **Federated Learning is a Lens towards a Democratized Future for the Scaling Law Era**  
H. H. Jiang, B. Askin, G. Joshi, C. Joe-Wong  
*ICML 2026 (Position Track)* [[Link](#)]

**Undergraduate Research (Bilkent, 2020–2022):** developed deep learning methods (transformers, CNNs, deep equilibrium models) for medical imaging, resulting in **2 IEEE Transactions on Medical Imaging papers** and **4 other conference/journal papers** (see [Google Scholar](#)).

## SELECTED COURSES AND RELEVANT SKILLS

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**CMU:** Ph.D.-level Optimization and Statistics courses, Generative AI, MDPs and RL, Deep RL

**Bilkent:** Linear Algebra, Stochastic Models, Signals and Systems, Digital Signal Processing

**Skills:** Python, PyTorch, Hugging Face Libraries, vLLM, VERL, OpenCV, Apache Spark

## PROFESSIONAL SERVICE

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- Reviewer for NeurIPS, ICLR, ICML, AISTATS, ISIT, AAAI, TMLR, IEEE/ACM ToN and MLSys
- Teaching Assistant for [Introduction to ML for Engineers \(18-661\)](#) and [Algorithms for Large-scale Distributed ML and Optimization \(18-667\)](#)

## SELECTED AWARDS AND ADDITIONAL INFORMATION

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- 2023-2024 Ben Cook Presidential Graduate Fellowship at CMU
- 2022-2023 Carnegie Institute of Technology Dean's Fellowship at CMU
- Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc.
- Turkish Government Merit Scholarship: Awarded stipend during the B.Sc.
- Ranked 252nd/2M in the university and 3rd/104k in the postgraduate entrance exams in Turkey
- Volunteer at Young Guru Academy (an international NGO based in Istanbul) from 2018 to 2020